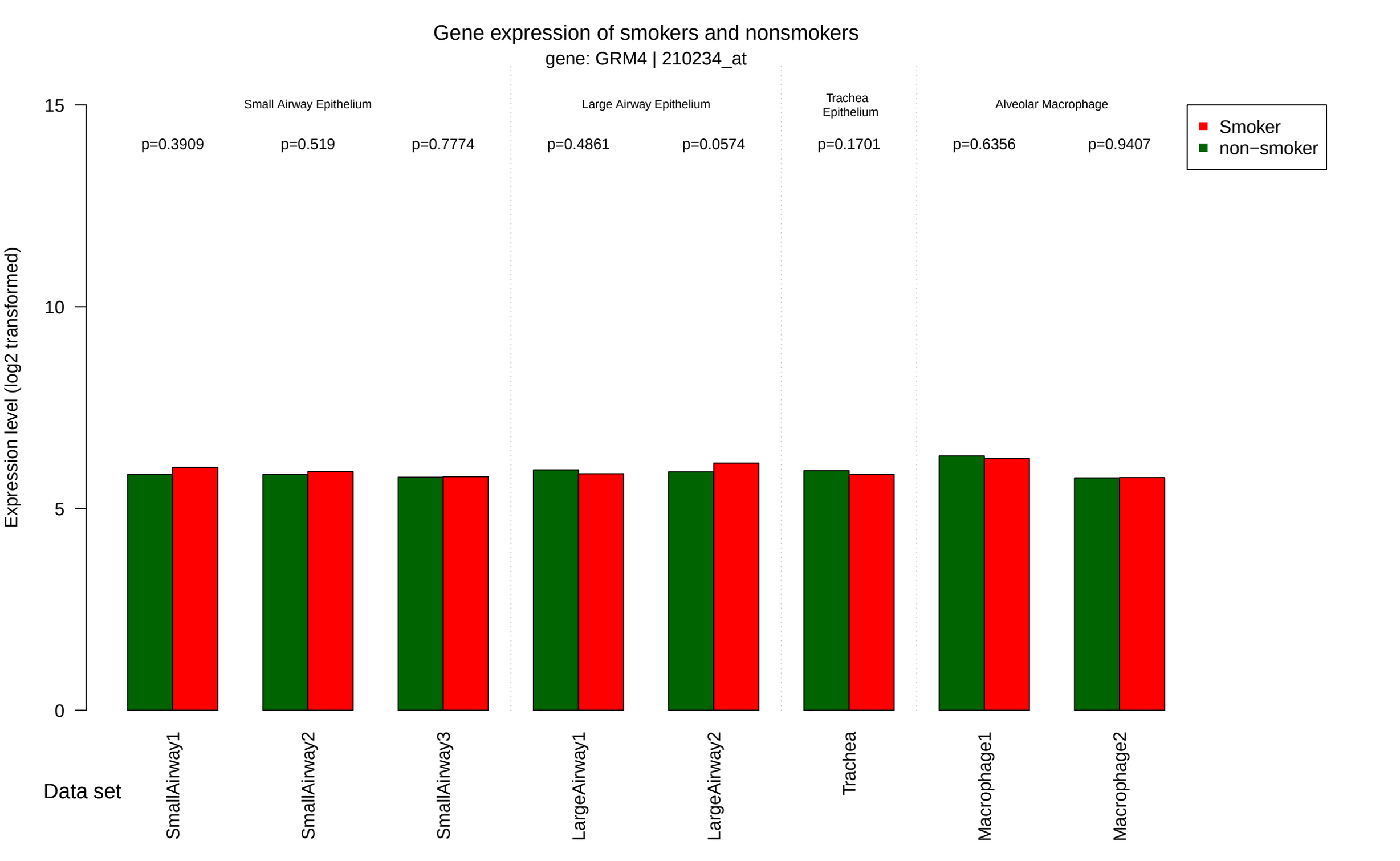
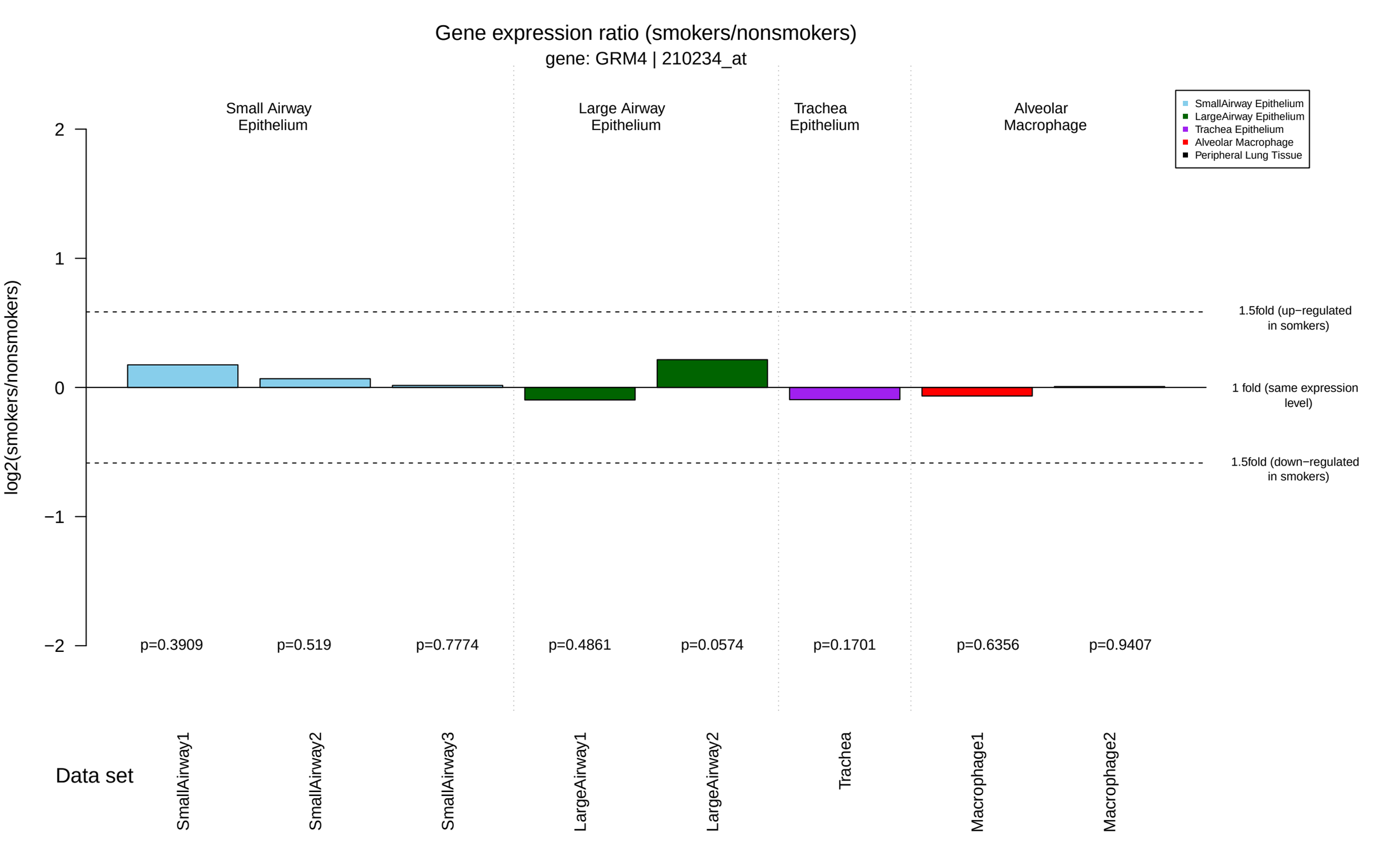
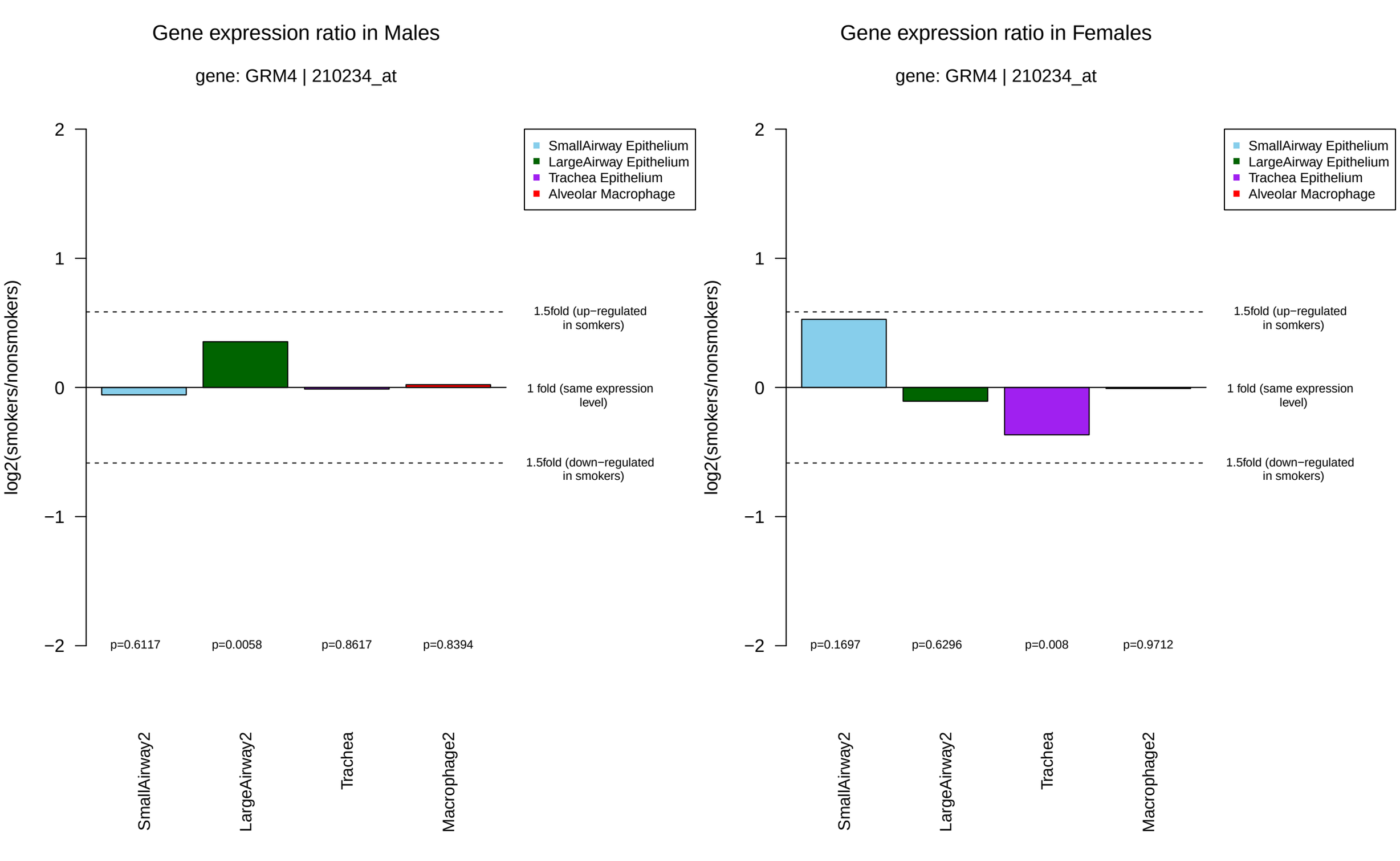
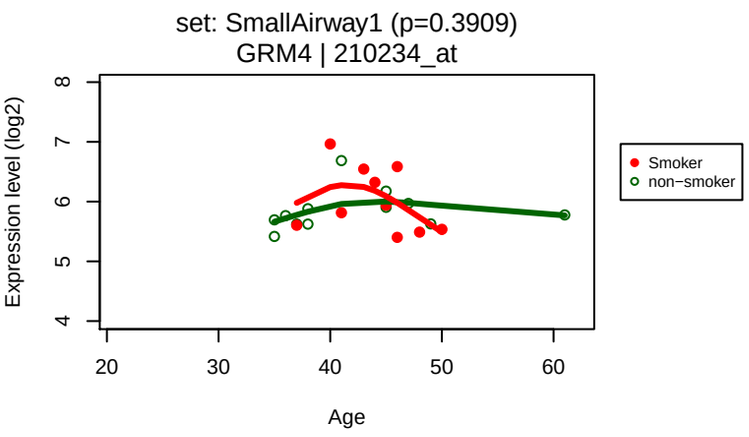
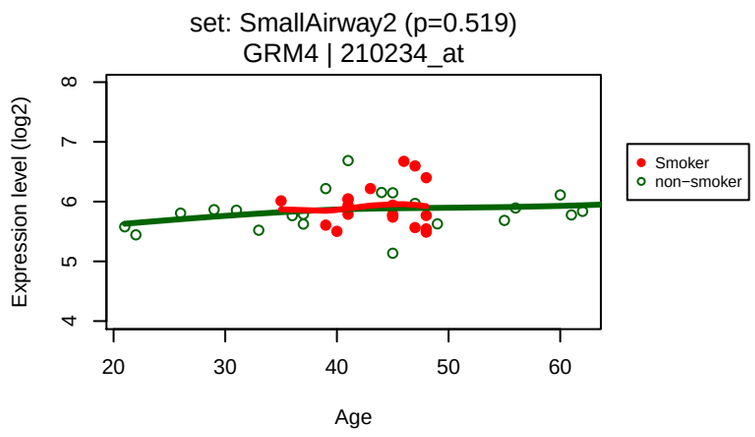
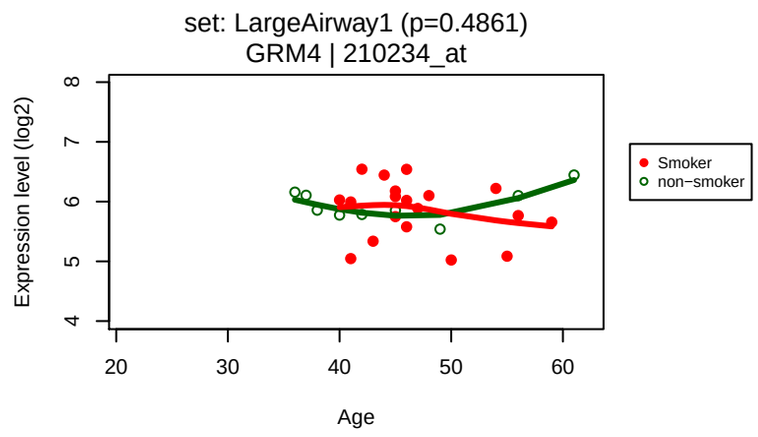

Gene expression of smokers and nonsmokers
gene: GRM4 | 210234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3909
p=0.519
p=0.7774
p=0.4861
p=0.0574
p=0.1701
p=0.6356
p=0.9407
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

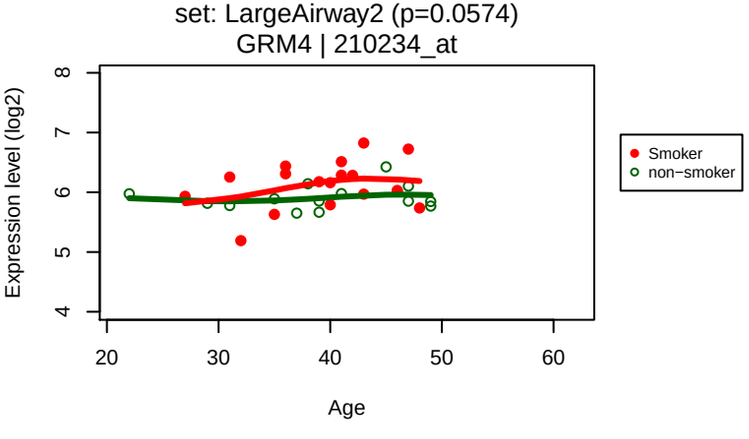

Gene expression ratio (smokers/nonsmokers)
gene: GRM4 | 210234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3909
p=0.519
p=0.7774
p=0.4861
p=0.0574
p=0.1701
p=0.6356
p=0.9407
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

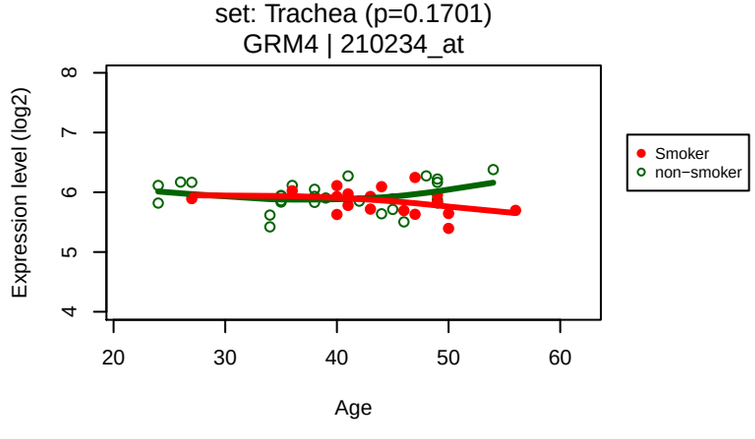

Gene expression ratio in Males
gene: GRM4 | 210234_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6117
p=0.0058
p=0.8617
p=0.8394
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRM4 | 210234_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1697
p=0.6296
p=0.008
p=0.9712
SmallAirway2
LargeAirway2
Trachea
Macrophage2

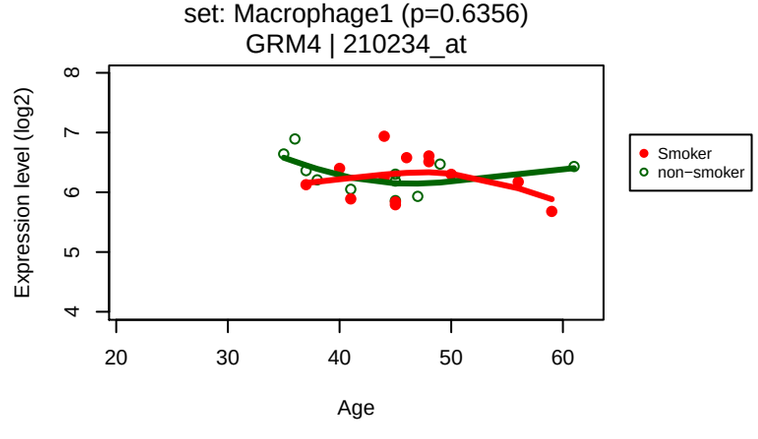

set: SmallAirway1 (p=0.3909)
GRM4 | 210234_at
Expression level (log2)
Age
Smoker
non−smoker

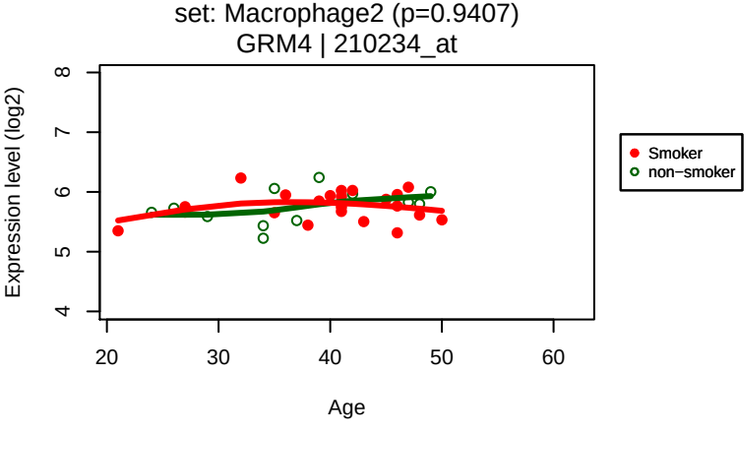

set: SmallAirway2 (p=0.519)
GRM4 | 210234_at
Expression level (log2)
Age
Smoker
non−smoker

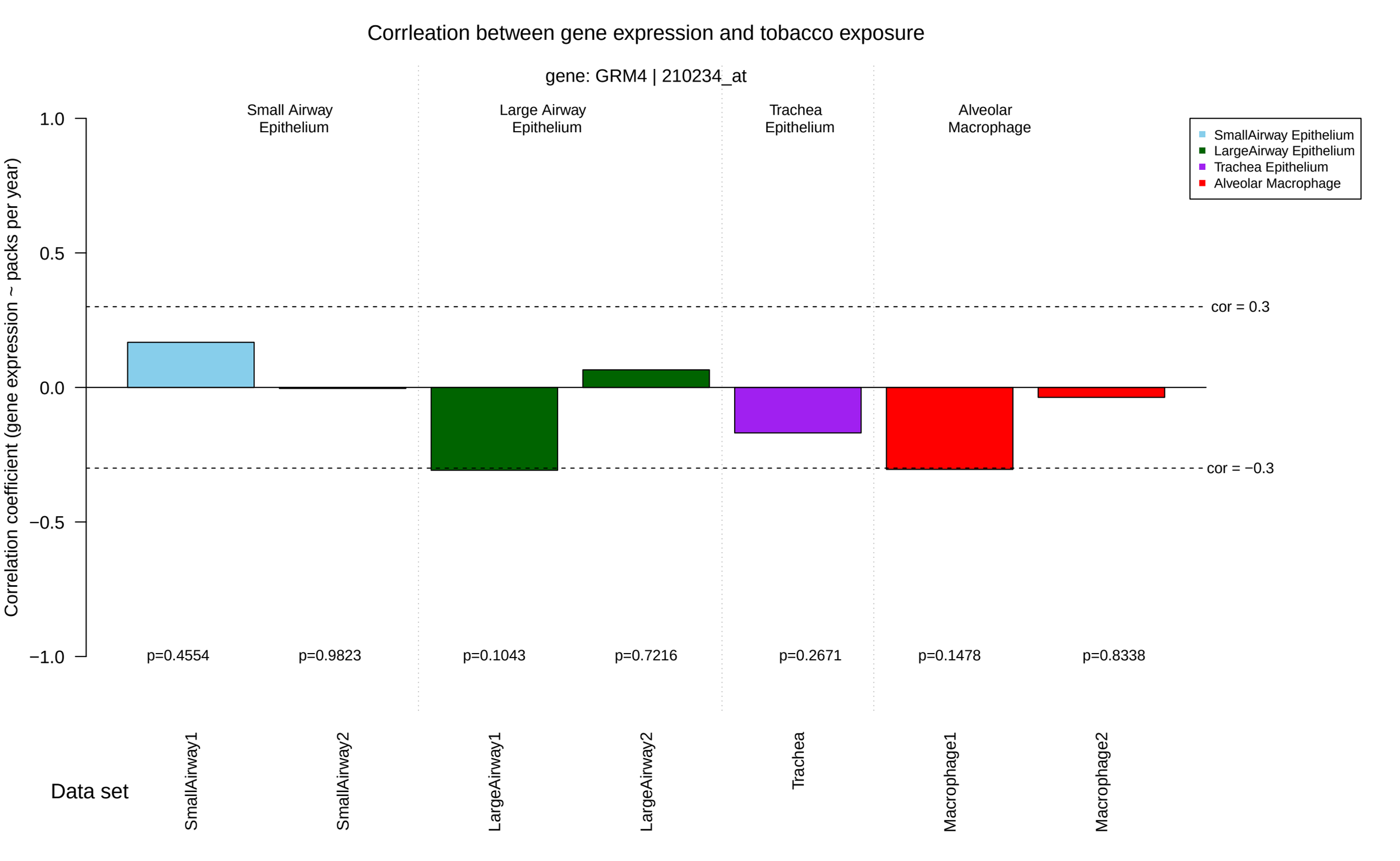

set: LargeAirway1 (p=0.4861)
GRM4 | 210234_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0574)
GRM4 | 210234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1701)
GRM4 | 210234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6356)
GRM4 | 210234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9407)
GRM4 | 210234_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRM4 | 210234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4554
p=0.9823
p=0.1043
p=0.7216
p=0.2671
p=0.1478
p=0.8338
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2